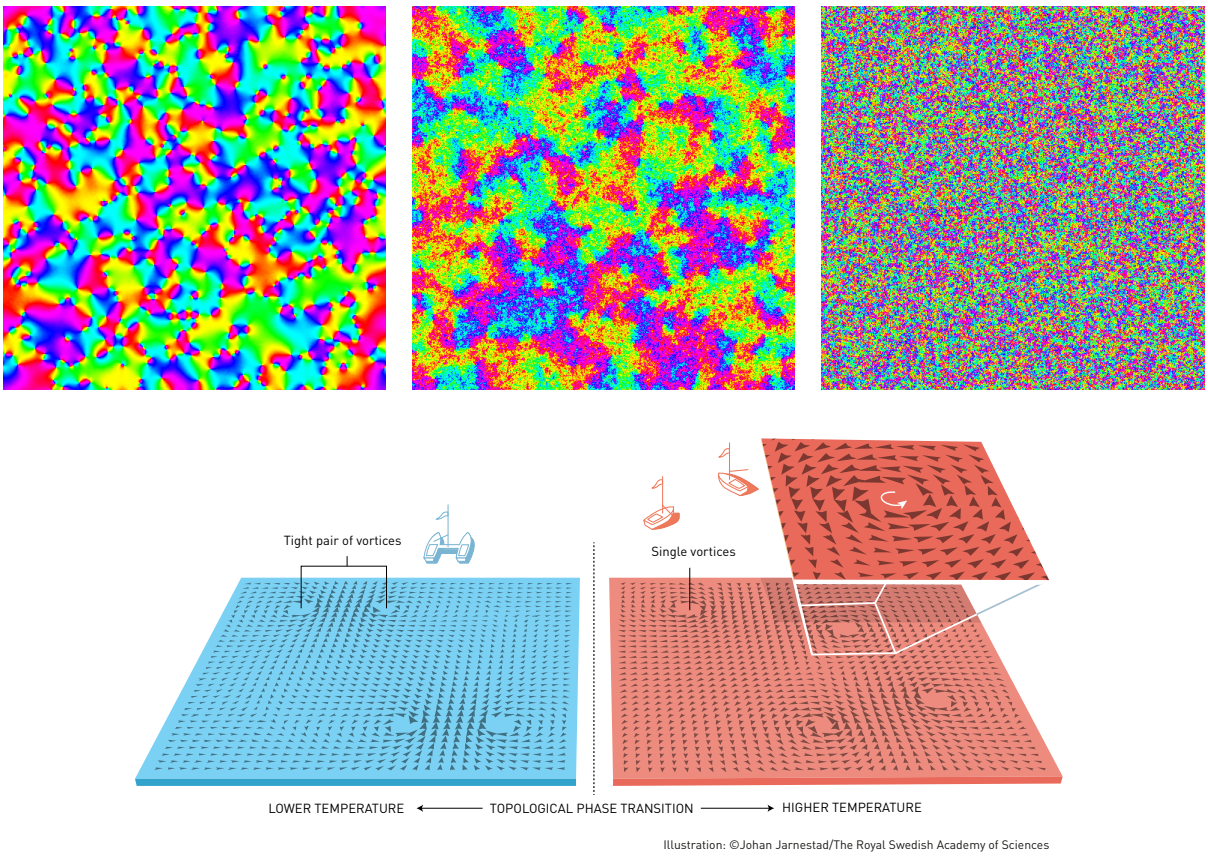


Figure 1: The XY model is the simplest model to study critical phenomena in the universality class characterized by 2-component order parameters. In 2D, this model has excitations with nontrivial topology which can separate or annihilate. At low temperature $T < T_c$, the system consists of bound vortex-antivortex pairs, whereas at $T > T_c$ these topological defects are liberated. The topological phase transition at $k_\mathrm{B}T_c/J \approx 0.9$ is of infinite order and is associated with the unbinding of vortex-antivortex pairs instead of the breaking of global $O(2)$ symmetry. This transition is known as the Brezinskii-Klosterlitz-Thouless transition (see The Nobel Prize in Physics 2016).

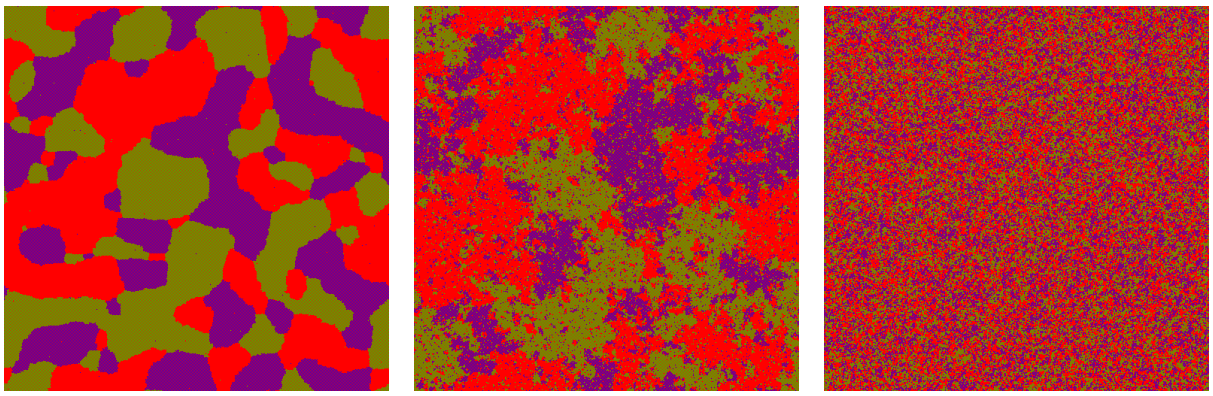


Figure 2: The $q = 3$ Potts model in 2D exhibits a second-order phase transition at $k_\mathrm{B}T_c/J = 1/\log(1+\sqrt{3})$.

Fluid system

We consider a fluid system described by the grand partition function $\Xi(z,V) = (1+z)^V(1+z^{\alpha V})$ where $\alpha > 0$ and V is the volume of the system in some units (e.g. the hard core radius) and z is the fugacity $\exp(\beta\mu)$.

(a) From the Lee-Yang theorem, we expect that there will be no phase transition at $z \neq 1$ but something interesting could happen at $z = 1$:

$$F_\infty(z) = \begin{cases} \log(1+z) & |z| < 1 \\ \log(1+z) + \alpha \log z & |z| > 1 \end{cases}$$

is continuous for all z, but is only analytic for $|z| \neq 1$. Now, we eliminate the dependence on z by using the fact that $pV = k_\mathrm{B}T \log \Xi$ and $N = z\,\partial_z \log \Xi$. The particle density n in the thermodynamic limit is given by

$$n = \begin{cases} \frac{z}{1+z} & |z| < 1 \\ \frac{z}{1+z} + \alpha & |z| > 1 \end{cases} .$$

Observe that there is a jump in particle density from $n = 1/2$ to $n = 1/2 + \alpha$ at $p = k_\mathrm{B}T \log 2$. This is a hint that tells us to expect a phase transition. The equation of state for $|z| < 1$ reads

$$p = k_\mathrm{B}T \log\left(\frac{1}{1-n}\right), \qquad n \in [0, 1/2) .$$

While for $|z| > 1$, the equation of state is given by

$$p = k_\mathrm{B}T \left[\log\left(\frac{1}{1+\alpha-n}\right) + \alpha \log\left(\frac{n-\alpha}{1+\alpha-n}\right)\right], \qquad n \in (1/2+\alpha, 1+\alpha) .$$

The plot of pressure p against the specific volume $v = 1/n$ is shown in Fig. 3. Observe that the curve is similar to the pressure-volume plot of the liquid-gas phase diagram under the coexistence curve[1]. This is a first-order phase transition! In this case, the specific volumes of the two phases at coexistence are $v_g = 2$ and $v_l = (1/2+\alpha)^{-1}$.

(b) The grand partition function Ξ has αV simple zeros at $z = e^{i(2k+1)\pi/\alpha V}$ where $k \in \{0, \pm 1, \ldots, \pm(\alpha V - 1)/2\}$. Note that when V is finite, the function Ξ is a finite polynomial of z that obeys $\overline{\Xi}(z) = \Xi(\bar{z})$, thus its zeros must either sit on the real negative axis or come in complex pairs. This symmetry is important, it leads all the zeros to lie exactly on the unit circle $|z| = 1$, uniformly distributed and separated by the arc length distance $2\pi/\alpha V$. In the thermodynamic limit, however, the maximum number of particles that can be fitted in the system diverges and Ξ is now defined as an infinite series. In this limit, the zeros coalesce and impinge the positive real axis at $z = 1$. Hence, there is a phase transition at $z = 1$. It is also worth noting that the density of the zeros and the manner in which the transition point is reached are generally related to the universality class of the transition. For instance, see Phys. Rev. Lett. **81**, 2000 (1998), Phys. Rev. Lett. **84**, 4794 (2000), and Phys. Rev. Lett. **118**, 180601 (2017).

[1] See p. 35 of Chapter 2 in PH7002 Lectures.

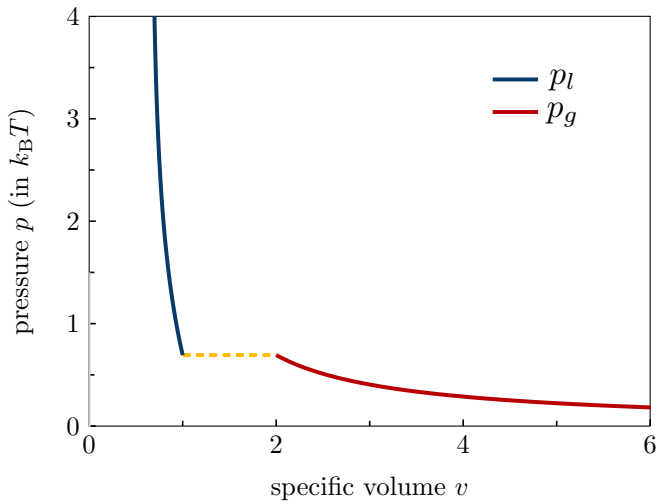


Figure 3: The pressure-volume plot obtained from the grand partition function Ξ at the thermodynamic limit with $\alpha = 1/2$. Dashed line represents the two-phase region $v \in [(1/2+\alpha)^{-1}, 2]$ at $p = k_{\mathrm{B}}T \log 2$.

(c) From part (a), we found that the equation of state in the gas phase is given by

$$p_g = k_{\mathrm{B}}T \log\left(\frac{v}{v-1}\right) .$$

Suppose we can consider the region beyond which the phase transition occurs, i.e. $v < 2$, observe that the equation of state is smooth throughout as if there is no phase transition at all! This is not surprising since there are no phase transitions in a finite system, so one has to take the limit $V \to \infty$ in order to see the discontinuity. Having done part (a), we know that the equation of state takes the form of p_g only for $v \in (2, \infty)$.

(d) We know from Fig. 3 that v is a monotonically decreasing function of p, so the isothermal compressibility is strictly positive. The grand partition function used in this question is therefore physically acceptable since the system obeys Le Chatelier's principle.

Yeomans 4.3

Consider the Hamiltonian of a spin model with uniform interaction length across its nearest neighbors and zero external field:

$$\mathcal{H} = -J \sum_{\langle ij \rangle} \vec{s}_i \cdot \vec{s}_j \ ,$$

where $\langle ij \rangle$ denotes summation over nearest neighbors. Using the mean-field approach, we choose the trial Hamiltonian whose states being different combinations of spins across $\{j\}$:

$$\mathcal{H}_0 = -\sum_j \vec{H}_{0,j} \cdot \vec{s}_j \quad \xrightarrow{\{\vec{H}_{0,j}\}=\vec{H}_0} \quad \mathcal{H}_0 = -\vec{H}_0 \cdot \sum_j \vec{s}_j \ .$$

In the trial Hamiltonian, we pretend that the spins are decoupled from each other and each of them experiences the *effective external field* $\vec{H}_{0,j}$. If we consider $\vec{H}_{0,j}$ to be isotropic i.e. $\vec{H}_{0,j} = \vec{H}_0$ for all sites, we can take it out from the summation and simplify $\mathcal{H}_0$. We work on the general case $\vec{H}_{0,j}$ for now and impose the isotropic later on. If the canonical partition function associated with $\mathcal{H}_0$ is given by $\mathcal{Z}_0$, the corresponding free energy reads $\mathcal{F}_0 = -k_\mathrm{B}T \log \mathcal{Z}_0$. The rigorous upper bound on the true free energy $\mathcal{F}$ is then given by the Bogoliubov inequality $\mathcal{F} \leq \mathcal{F}_0 + \langle \mathcal{H} - \mathcal{H}_0 \rangle_0$ with

$$\langle \mathcal{H} - \mathcal{H}_0 \rangle_0 = \frac{\sum\limits_{\{\vec{s}\}} \left(-J \sum\limits_{\langle ij \rangle} \vec{s}_i \cdot \vec{s}_j + \sum\limits_j \vec{H}_{0,j} \cdot \vec{s}_j \right) e^{\beta \cdot \sum\limits_i \vec{H}_{0,i} \cdot \vec{s}_i}}{\sum\limits_{\{\vec{s}\}} e^{\beta \sum\limits_i \vec{H}_{0,i} \cdot \vec{s}_i}} = -J \sum_{\langle ij \rangle} \langle \vec{s}_i \cdot \vec{s}_j \rangle_0 + \sum_j \langle \vec{H}_{0,j} \cdot \vec{s}_j \rangle_0 \ .$$

where we've taken the thermal average using weights from the trial ensemble (denoted by the subscript 0). Now, we want to find the value of $\vec{H}_{0,k}$ such that $\mathcal{H}_0$ provides the finest approximation to $\mathcal{H}$. From the variational principle, we know that $\vec{H}_{0,k}$ is also the one that minimizes the free energy $\mathcal{F}$. Taking the first derivative of $\mathcal{F}$ with respect to the variational parameters $\vec{H}_{0,k}$ to be zero, we have the condition

$$\frac{\partial \mathcal{F}_0}{\partial \vec{H}_{0,k}} - J \frac{\partial}{\partial \vec{H}_{0,k}} \left[\sum_{\langle ij \rangle} \langle \vec{s}_i \cdot \vec{s}_j \rangle_0 \right] + \frac{\partial}{\partial \vec{H}_{0,k}} \left[\sum_j \vec{H}_{0,j} \cdot \langle \vec{s}_j \rangle_0 \right] = 0 \ .$$

Let's evaluate these terms separately. Since the partition function is non-interacting, the first term reads

$$\frac{\partial \mathcal{F}_0}{\partial \vec{H}_{0,k}} = -\frac{1}{\beta} \frac{\partial}{\partial \vec{H}_{0,k}} \log Z_0 = -\frac{1}{\beta} \frac{\partial}{\partial \vec{H}_{0,k}} \log \left(\sum_{\{\vec{s}\}} e^{\beta \sum\limits_i \vec{H}_{0,i} \cdot \vec{s}_i} \right) = -\frac{\sum\limits_{\{\vec{s}\}} \vec{s}_k \, e^{\beta \sum\limits_i \vec{H}_{0,i} \cdot \vec{s}_i}}{\sum\limits_{\{\vec{s}\}} e^{\beta \sum\limits_i \vec{H}_{0,i} \cdot \vec{s}_i}} = -\langle \vec{s}_k \rangle_0 \ .$$

To get the mean-field approximation, we assume that all spins are uncorrelated[2], i.e. $\langle \vec{s}_i \cdot \vec{s}_j \rangle_0 = \langle \vec{s}_i \rangle_0 \cdot \langle \vec{s}_j \rangle_0$. The second term reduces to

$$-J \frac{\partial}{\partial \vec{H}_{0,k}} \left[\sum_{\langle ij \rangle} \langle \vec{s}_i \cdot \vec{s}_j \rangle_0 \right] = -J \sum_{\langle ij \rangle} \left(\frac{\partial \langle \vec{s}_i \rangle_0}{\partial \vec{H}_{0,k}} \cdot \langle \vec{s}_j \rangle_0 + \langle \vec{s}_i \rangle_0 \cdot \frac{\partial \langle \vec{s}_j \rangle_0}{\partial \vec{H}_{0,k}} \right) = -J \sum_{i \in nn(k)} \left(\langle \vec{s}_i \rangle_0 \cdot \frac{\partial \langle \vec{s}_k \rangle_0}{\partial \vec{H}_{0,k}} \right) \ .$$

In the last step, we've simplified the $\langle ij \rangle$ summation using the fact the derivative only acts on terms involving spin k and that there is one term for each nearest-neighbors (nn) of site k resulting in a sum over i. Now,

$$\frac{\partial}{\partial \vec{H}_{0,k}} \left[\sum_j \vec{H}_{0,j} \cdot \langle \vec{s}_j \rangle_0 \right] = \langle \vec{s}_k \rangle + \vec{H}_{0,k} \cdot \frac{\partial \langle \vec{s}_k \rangle_0}{\partial \vec{H}_{0,k}} \ .$$

[2]See Chapter 2.2 in Statistical Physics of Fields (2007) by M. Kardar.

Collecting the terms, we find

$$-\langle \vec{s}_k \rangle_0 - J \sum_{i \in nn(k)} \left(\langle \vec{s}_i \rangle_0 \cdot \frac{\partial \langle \vec{s}_k \rangle_0}{\partial \vec{H}_{0,k}} \right) + \langle \vec{s}_k \rangle_0 + \vec{H}_{0,k} \cdot \frac{\partial \langle \vec{s}_k \rangle_0}{\partial \vec{H}_{0,k}} = 0$$

$$\left(-J \sum_{i \in nn(k)} \langle \vec{s}_i \rangle_0 + \vec{H}_{0,k} \right) \cdot \frac{\partial \langle \vec{s}_k \rangle_0}{\partial \vec{H}_{0,k}} = 0$$

We can further simplify the above equation by imposing the isotropic condition, i.e. $\langle \vec{s}_i \rangle_0 = \langle \vec{s}_j \rangle_0 \equiv \langle \vec{s} \rangle_0$ and $\vec{H}_{0,i} = \vec{H}_{0,j} = \vec{H}_0$. Hence, there is no longer explicit i-dependence inside the summation and the inner sum $\sum_{i \in \mathrm{nn}(k)}$ simply reduces to z, where the coordination number z is equal to the number of neighbors of any given site. The value of $\vec{H}_0$ that minimizes $\mathcal{F}$ is then

$$\vec{H}_0 = Jz \langle \vec{s} \rangle_0 \ .$$

Substituting $\vec{H}_0$ into the free energy $\mathcal{F}$ gives

$$\mathcal{F}_{\mathrm{MF}} = \min_{\vec{H}_0} \mathcal{F} = -k_{\mathrm{B}} T \log \mathcal{Z}_0 - \frac{J}{2} \sum_{i=1}^{N} \sum_{j \in \mathrm{nn}(i)} \langle \vec{s} \rangle_0^2 + Jz \sum_{j=1}^{N} \langle \vec{s} \rangle_0^2 = -k_{\mathrm{B}} T \log \mathcal{Z}_0 + \frac{JNz}{2} \langle \vec{s} \rangle_0^2 \ .$$

The factor $1/2$ comes out from $\langle ij \rangle$ summation to avoid double counting pairs of sites and $Nz/2$ thus signifies the total number of bonds in the system. This variational bound is remarkably useful for practical calculations away from critical points!

Remarks: The expression of $\mathcal{F}_{\mathrm{MF}}$ obtained in this question will be particularly helpful when we evaluate the mean-field free energies of the three-state Potts model (Yeomans 4.4) and the XY Model.

Yeomans 4.4

The main objective of this question is to derive mean-field equations for the three-state Potts model using the result obtained in the previous question. The Potts model is defined by the Hamiltonian

$$\mathcal{H}_{\mathrm{Potts}} = -J_{\mathrm{Potts}} \sum_{\langle ij \rangle} \delta_{\sigma_i \sigma_j} \ ,$$

where $\delta_{\sigma_i \sigma_j}$ is the usual Kronocker delta, which equals 1 if $\sigma_i = \sigma_j$ and zero otherwise.

(a) Now consider the Hamiltonian

$$\mathcal{H} = -J \sum_{\langle ij \rangle} \vec{s}_i \cdot \vec{s}_j \ ,$$

with

$$\{\vec{s}\} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \ \frac{1}{2} \begin{pmatrix} -1 \\ \sqrt{3} \end{pmatrix}, \ -\frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \right\} \ .$$

Notice that $\vec{s}_i \cdot \vec{s}_j = 1$ if $i = j$ while $\vec{s}_i \cdot \vec{s}_j = -1/2$ if $i \neq j$ from which we obtain the relation

$\vec{s}_i \cdot \vec{s}_j = (3\delta_{\sigma_i\sigma_j} - 1)/2$. And since the Hamiltonian sums over all nearest-neighbor pairs, the relation applies to the full many-spin system. We find

$$\mathcal{H}_{\text{Potts}} = \mathcal{H} + \frac{JNz}{4} \ .$$

for $J_{\text{Potts}} = 3J/2$, which means that the three-state Potts model described by $\mathcal{H}_{\text{Potts}}$ with spin-spin interaction $J_{\text{Potts}} = 3J/2$ is physically equivalent to a spin model described by $\mathcal{H}$ with interaction J, also known as the *Heisenberg model*[3]. Note that the constant term $JNz/4$ is irrelevant since it doesn't change the probability of the energy states with configuration $\{\sigma_i\}$.

(b) From now on, we set $J_{\text{Potts}} \to 3J/2$ so that $\mathcal{H}_{\text{Potts}} = \mathcal{H}$. The partition function $\mathcal{Z}_0$ of trial Hamiltonian $\mathcal{H}_0$ reads

$$\mathcal{Z}_0 = \operatorname{Tr} e^{-\beta\mathcal{H}_0} = \sum_{\{\vec{s}\}} e^{\beta\vec{H}_0\cdot\sum_i \vec{s}_i} = \left(\sum_{\vec{s}} e^{\beta\vec{H}_0\cdot\vec{s}_1}\right)\left(\sum_{\vec{s}} e^{\beta\vec{H}_0\cdot\vec{s}_2}\right)\cdots\left(\sum_{\vec{s}} e^{\beta\vec{H}_0\cdot\vec{s}_N}\right) \ ,$$

where $\vec{H}_0 = (H_0, H_0')$ and for a given k,

$$\sum_{\vec{s}} e^{\beta\vec{H}_0\cdot\vec{s}_k} = e^{\beta H_0} + e^{\beta\left(-H_0/2+\sqrt{3}\,H_0'/2\right)} + e^{\beta\left(-H_0/2-\sqrt{3}\,H_0'/2\right)} \ .$$

If we focus on the ordered state where $(1,0)$ predominates, the above expression reduces to

$$\sum_{\vec{s}} e^{\beta\vec{H}_0\cdot\vec{s}_k} = e^{\beta H_0} + 2e^{-\beta H_0/2} \ .$$

With isotropic condition $\vec{s}_i = \vec{s}_j \equiv \vec{s}$, we can evalute the temperature-dependent $\langle\vec{s}\rangle_0$ via

$$\langle\vec{s}\rangle_0 = -\frac{\partial\mathcal{F}_0}{\partial\vec{H}_0} = \frac{1}{\beta}\left(\frac{\partial}{\partial H_0}, \frac{\partial}{\partial H_0'}\right)\log Z_0 = \frac{1}{\beta}\left(\frac{\partial}{\partial H_0}, 0\right)\log Z_0 \ .$$

From **Yeomans 4.3**, we know that $\langle\vec{s}\rangle_0 = \vec{H}_0/Jz$. The non-vanishing component of $\langle\vec{s}\rangle_0$ then reads

$$\frac{H_0}{Jz} = \frac{\exp(3\beta H_0/2) - 1}{\exp(3\beta H_0/2) + 2} \ .$$

And the mean-field free energy is given by

$$\mathcal{F}_{\text{MF}} = \mathcal{F}_0 + \langle\mathcal{H}_{\text{Potts}} - \mathcal{H}_0\rangle_0 = -Nk_{\text{B}}T\log\left(\exp(\beta H_0) + 2\exp(-\beta H_0/2)\right) + \frac{NH_0^2}{2Jz} \ .$$

(c) From this point forward, we denote $\langle\vec{s}\rangle_0$ as s. Expanding the free energy up to s^4 and ignoring constant term :

$$\frac{1}{N}\mathcal{F}_{\text{MF}} = -\frac{1}{4}Jz(\beta Jz - 2)s^2 - \frac{1}{24}\beta^2(Jz)^3 s^3 + \frac{1}{64}\beta^3(Jz)^4 s^4 \ .$$

The cubic term is nonzero, so we expect a first-order transition.

[3]See Chapter 6 in Statistical Physics of Fields (2007) by M. Kardar.

(d) The averaged spin has to be self-consistent, and it should satisfy

$$s = \frac{e^{(3\beta s J z)/2} - 1}{e^{(3\beta s J z)/2} + 1} \ .$$

It is easy to check that $s = 1/2$ satisfy the equation above. Could it be the other minimum of $\mathcal{F}_{\rm MF}$ as $T \to T_c$? Using T_c, we have $\mathcal{F}_{\rm MF}(s=0) = \mathcal{F}_{\rm MF}(s=1/2) = -N(3\log 3)/(8\log 2)$. To verify that indeed $s = 1/2$ is a minimum, we use

$$\frac{1}{NJz}\frac{\partial \mathcal{F}_{\rm MF}}{\partial s} = -1 + s + \frac{3\, e^{-3\beta s J z/2}}{1 + 2\, e^{-3\beta s J z/2}} \ .$$

This yields zero for $s = 1/2$ and $s = 0$. The jump in the order parameter (i.e. the magnetization) at Tc is then given by $Jz\Delta s = Jz/2$.

XY Model

So far we've seen n-vector spin models with $n = 1$ (Ising model) and $n = 3$ (three-state Potts model, Yeomans 4.4). In this question, we'll deal with a spin model associated to $n = 2$, which is also known as the XY model. Consider a d-dimensional lattice model in which the spin variables at each site j are free to rotate within the x-y plane, i.e. $\vec{s}_j = (\cos\theta_j, \sin\theta_j)$ where $0 \le \theta_j < 2\pi$ is the angle that the spin $\vec{s}_j$ makes with some fixed reference direction. The model can be described by the Hamiltonian

$$\mathcal{H} = -J\sum_{\langle ij\rangle} \vec{s}_i \cdot \vec{s}_j = -J\sum_{\langle ij\rangle} \cos(\theta_i - \theta_j) \ .$$

Now consider a trial Hamiltonian

$$\mathcal{H}_0 = -\vec{H}_0 \cdot \sum_i \vec{s}_i \ .$$

Note that we can choose $\vec{H}_0$ and $\langle \vec{s}\rangle$ to point along the x-axis since $\mathcal{H}$ possesses spin-rotational invariance about the z-axis. From Yeomans 4.3, the effective field is given by $\vec{H}_0 = 2dJ\langle \vec{s}\rangle_0$, where we've used the fact that $z = 2d$. Let the non-vanishing component of $\langle \vec{s}\rangle$ be $\langle s\rangle \equiv \langle \cos\theta\rangle_0$, we have $\vec{H}_0 = (2dJ\langle s\rangle, 0)$. The mean-field energy then reads

$$\mathcal{F}_{\rm MF} = -\frac{1}{\beta}\log\left(\prod_i \int \mathcal{D}[\theta]\, e^{2d\beta J\langle s\rangle\cos\theta_i}\right) + NdJ\langle s\rangle^2 = -\frac{N}{\beta}\log\left(\int \frac{d\theta}{2\pi}\, e^{2d\beta J\langle s\rangle\cos\theta}\right) + NdJ\langle s\rangle^2 \ ,$$

where the measure $\mathcal{D}[\theta]$ ensures that we integrate over all possible configurations of θ since the spins are free to rotate in all directions. We can write $\mathcal{F}_{\rm MF}/N$ in terms of modified Bessel function $I_\nu(z)$:

$$\frac{1}{N}\mathcal{F}_{\rm MF} = -\frac{1}{\beta}\log I_0(2d\beta J\langle s\rangle) + dJ\langle s\rangle^2 \ .$$

The thermal average magnetization $\langle \vec{s}\rangle_0$ per spin is

$$\langle \vec{s}\rangle_0 = (\langle s\rangle, 0) = \left(\frac{1}{\beta}\frac{\partial}{\partial(2dJ\langle s\rangle)}\log I_0(2d\beta J\langle s\rangle), 0\right) = \left(\frac{I_1(2d\beta J\langle s\rangle)}{I_0(2d\beta J\langle s\rangle)}, 0\right) \ .$$

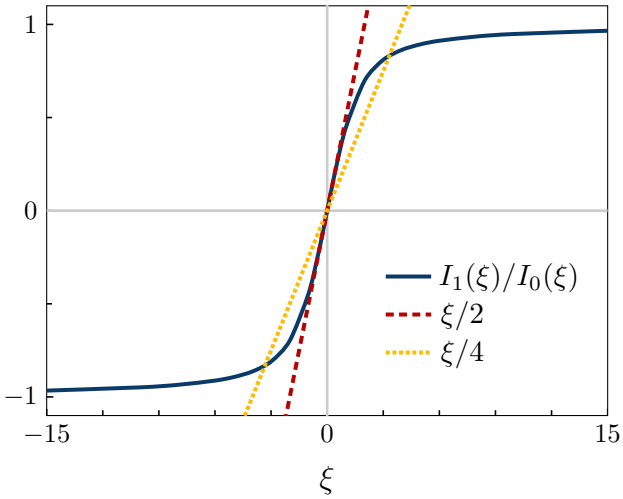
from which we obtain the self-consistency equation

$$\langle s\rangle = \frac{I_1(2d\beta J\langle s\rangle)}{I_0(2d\beta J\langle s\rangle)} \ .$$

Let $\xi = 2d\beta J\langle s\rangle$, with the given expansion of modified Bessel function $I_\nu(z)$ we have at small ξ :

$$\langle s\rangle = \frac{1}{2}\xi - \frac{1}{16}\xi^3 + \mathcal{O}(\xi^5)$$

and at large ξ, we find that $\langle s\rangle = 1 - \mathcal{O}(\xi^{-1})$.



The behavior of the self-consistency equation is similar to that of Ising model and it is therefore straightforward for us to get the critical (inverse) temperature $\beta_c = 1/dJ$. Expanding $\langle s\rangle$ at $T \approx T_c$ then gives

$$\langle s\rangle = \sqrt{2\left(1 - \frac{T}{T_c}\right)}$$

Remarks: Observe that the critical temperature that we obtained is far from the correct value $k_\mathrm{B}T_c/J \approx 0.9$. In mean-field theory, we've ignored thermal fluctuations by minimizing the free energy rather than sampling it with a thermal distribution. Near T_c, however, there are fluctuations on all length scales from whose effects the true critical exponents emerges. The mean-field theory would give quantitatively more accurate results in higher dimensions, as dictated by the *Ginzburg criterion*[4]. The true value of T_c can be determined by various methods such as measuring the properties of the model at different temperatures via Monte Carlo simulations. For example, the spin-spin correlation function and susceptibility in Phys. Rev. B **20**, 3761 (1979), and helicity modulus in Phys. Rev. B **37**, 5986(R) (1998).

[4]See Chapter 3.7 in Statistical Physics of Fields (2007) by M. Kardar.

Yeomans 4.4

$$\mathcal{H} = -J \sum_{\langle ij \rangle} \delta_{\sigma_i \sigma_j} \qquad \sigma_i = 1,2,3$$

3 state Potts model

(i) $$\mathcal{H} = -J \sum_{\langle ij \rangle} \vec{S}_i \cdot \vec{S}_j \qquad \vec{S}_i = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1/2 \\ \sqrt{3}/2 \end{pmatrix}, \begin{pmatrix} -1/2 \\ -\sqrt{3}/2 \end{pmatrix}$$

look at

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \qquad \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -1/2 \\ \sqrt{3}/2 \end{pmatrix} = -1/2 \qquad \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -1/2 \\ \sqrt{3}/2 \end{pmatrix} = -1/2$$

$$\begin{pmatrix} -1/2 \\ \sqrt{3}/2 \end{pmatrix} \cdot \begin{pmatrix} -1/2 \\ \sqrt{3}/2 \end{pmatrix} = 1 \qquad \begin{pmatrix} -1/2 \\ \sqrt{3}/2 \end{pmatrix} \cdot \begin{pmatrix} -1/2 \\ -\sqrt{3}/2 \end{pmatrix} = -1/2$$

$$\begin{pmatrix} -1/2 \\ -\sqrt{3}/2 \end{pmatrix} \cdot \begin{pmatrix} -1/2 \\ -\sqrt{3}/2 \end{pmatrix} = 1$$

We see that when $\vec{S}_i = \vec{S}_j \Rightarrow 1$

$\vec{S}_i \neq \vec{S}_j \Rightarrow -1/2$

Similarly to $\delta_{\sigma_i \sigma_j} = \begin{cases} 1 & , \sigma_i = \sigma_j \\ 0 & , \sigma_i \neq \sigma_j \end{cases}$

So the 2 expressions are equivalent up to <u>irrelevant</u> constant.

(ii) $\vec{H}_0 = \begin{pmatrix} H_0 \\ H_0' \end{pmatrix}$ $\qquad \mathcal{H}_0 = -\vec{H}_0 \cdot \sum_i' \vec{S}_i$

Mean Field equations from $\boxed{4.3}$

$$\vec{H}_0 = Jz \langle \vec{S} \rangle_0$$

$$F_{MF} = -k_B T \log Z_0 + \frac{Jz}{2} N \langle \vec{S} \rangle_0^2$$

$$\langle \vec{S} \rangle_0 = \frac{\mathrm{Tr}\left(\vec{S}\, e^{+\beta \vec{H}_0 \cdot \vec{S}} \right)}{\mathrm{Tr}\left(e^{+\beta \vec{H}_0 \cdot \vec{S}} \right)}$$ ← sum over $\vec{S} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1/2 \\ +\sqrt{3}/2 \end{pmatrix}, \begin{pmatrix} -1/2 \\ -\sqrt{3}/2 \end{pmatrix}$

set $H_0' = 0$ ($\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ ~~pot~~ dominated regime)

$$\vec{H}_0 \cdot \vec{S} = \begin{cases} \begin{pmatrix} H_0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = H_0 \\ \begin{pmatrix} H_0 \\ 0 \end{pmatrix} \begin{pmatrix} -1/2 \\ \pm\sqrt{3}/2 \end{pmatrix} = -\frac{H_0}{2} \end{cases} \Rightarrow \mathrm{Tr}\, e^{\beta \vec{H}_0 \cdot \vec{S}} = e^{\beta H_0} + 2 e^{-\beta \frac{H_0}{2}}$$

Likewise

$$\mathrm{Tr}\left(\vec{S}\, e^{\beta \vec{H}_0 \cdot \vec{S}} \right) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{\beta H_0} + \begin{pmatrix} -1/2 \\ \sqrt{3}/2 \end{pmatrix} e^{-\beta \frac{H_0}{2}} + \begin{pmatrix} -1/2 \\ -\sqrt{3}/2 \end{pmatrix} e^{-\beta H_0/2}$$

$$= \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{\beta H_0} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} e^{-\beta \frac{H_0}{2}}$$

From mean-field equation

$$\vec{H}_0 = Jz \langle \vec{S} \rangle_0$$

we find that

$$\begin{pmatrix} H_0 \\ 0 \end{pmatrix} = Jz \underbrace{\frac{\begin{pmatrix}1\\0\end{pmatrix} e^{\beta H_0} + \begin{pmatrix}-1\\0\end{pmatrix} e^{-\beta \frac{H_0}{2}}}{e^{\beta H_0} + 2 e^{-\beta \frac{H_0}{2}}}}_{\langle \vec{s} \rangle}$$

$$\Rightarrow \quad \frac{H_0}{Jz} = \frac{e^{\beta H_0} - e^{-\beta \frac{H_0}{2}}}{e^{\beta H_0} + 2 e^{-\beta \frac{H_0}{2}}} = \frac{e^{\frac{3}{2}\beta H_0} - 1}{e^{\frac{3}{2}\beta H_0} + 2}$$

The mean-field free energy is

$$\langle s \rangle_0 = \frac{H_0}{Jz}$$

$$Z_0 = \left(\mathrm{Tr}\, e^{-\beta \vec{H}_0 \cdot \vec{s}} \right)^N$$

$$F = -k_B T \log Z_0 + \frac{JzN}{2} \langle s \rangle_0^2$$

$$= -N k_B T \log \left(e^{\beta H_0} + 2 e^{-\beta \frac{H_0}{2}} \right) + \frac{NJz}{2} \left(\frac{H_0}{Jz} \right)^2$$

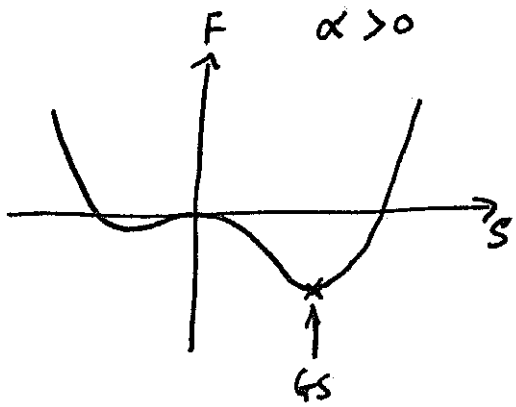
(iii) $$F = -N k_B T \log \left[e^{\beta J z \langle s \rangle_0} + 2 e^{-\frac{\beta}{2} J z \langle s \rangle_0} \right] + \frac{NJz}{2} \langle s \rangle_0^2$$

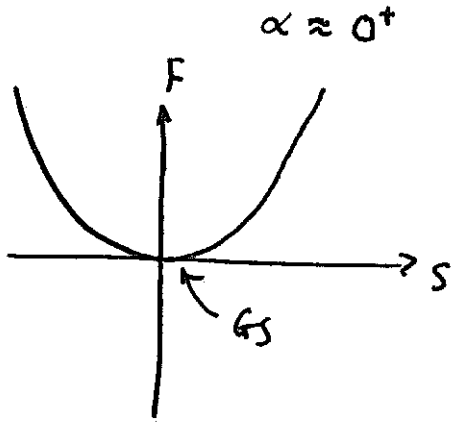
let $\alpha = \beta J z$ and $s = \langle s \rangle_0$

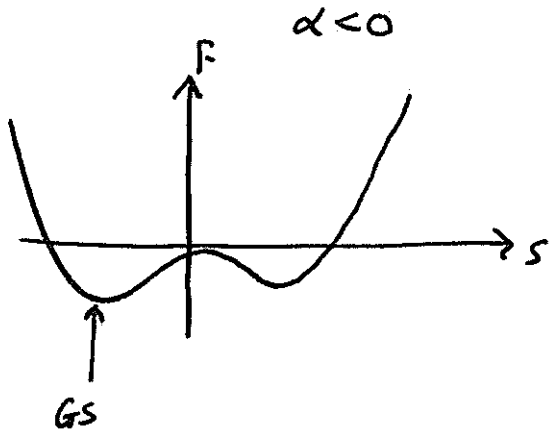
$$\Rightarrow \quad F = -N k_B T \left[\underbrace{\log e^{\alpha s}}_{\alpha s} + \log \left(1 + 2 e^{-\frac{3}{2}\alpha s} \right) - \frac{\alpha}{2} s^2 \right]$$

$$\simeq -N k_B T \left\{ \log 3 + \frac{1}{4}\alpha(\alpha - 2) + \frac{\alpha^3}{24} s^3 - \frac{\alpha^4}{64} s^4 + \cdots \right\}$$

cubic term







jump in location of $\langle s \rangle_{GS}$ $\Rightarrow$ 1st order P.T.

(iv)
$$\langle s \rangle_0 = \frac{e^{\frac{3}{2}\alpha\langle s\rangle_0} - 1}{e^{\frac{3}{2}\alpha\langle s\rangle_0} + 2} = f(\langle s \rangle_0)$$

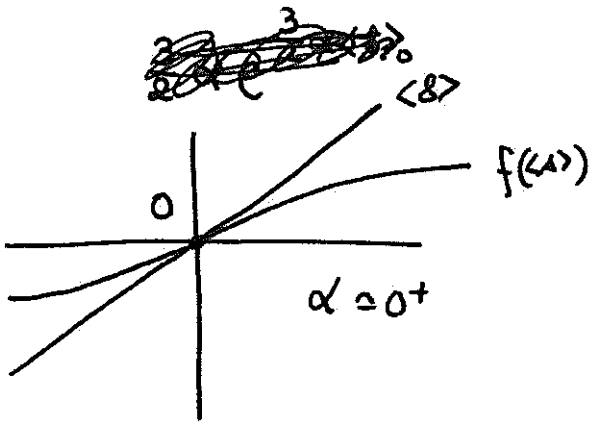
Find $f'(\langle s \rangle_0)\Big|_{\alpha_c} = 1$ $\qquad \alpha = \beta J z$

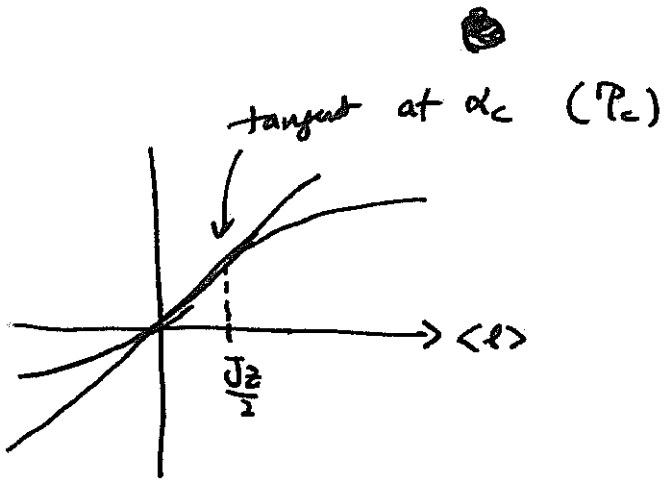
$\Rightarrow$
$$f'(\langle s \rangle_0) = \frac{\frac{3}{2}\alpha\, e^{\frac{3}{2}\alpha\langle s\rangle_0}}{e^{\frac{3}{2}\alpha\langle s\rangle_0} + 2} - \frac{(e^{\frac{3}{2}\alpha\langle s\rangle_0} - 1)}{(e^{\frac{3}{2}\alpha\langle s\rangle_0} + 2)^2} \cdot \frac{3}{2}\alpha\, e^{\frac{3}{2}\alpha\langle s\rangle_0}$$

At $\beta_c J z = \alpha_c = \frac{8}{3}\log 2$

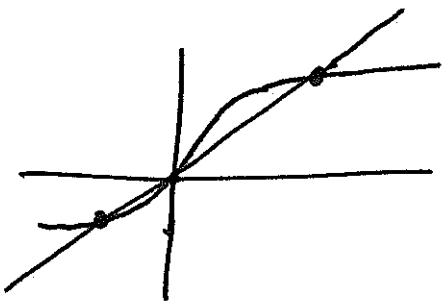
$$\frac{3}{2}\alpha_c = 4\log 2$$

We ~~see~~ that





$\alpha > \alpha_c$ $(T < T_c)$



jumps to

$$\langle s \rangle_0 \Rightarrow 0 \rightsquigarrow \tfrac{1}{2}$$

$$\frac{e^{\frac{1}{2}\cdot 4\log 2} - 1}{e^{\frac{1}{2}\cdot 4\log 2} + 2} = \frac{e^{\log 4} - 1}{e^{\log 4} + 2} = \frac{4-1}{4+2} = \tfrac{1}{2}$$

We see that if $\langle s \rangle_0 = 0 \rightsquigarrow \tfrac{1}{2}$ and if

$$\beta_c J z = \frac{8}{3}\log 2$$

then ~~tangent~~ this solution satisfies the mean field equation

$$\langle s \rangle_0 = \frac{e^{\frac{3}{2}\beta J z \langle s \rangle_0} - 1}{e^{\frac{3}{2}\beta J z \langle s \rangle_0} + 2}$$